Thao H. Nguyen

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Education

2018–2024 **Brown University**, Ph.D. in Computer Science.
- Dissertation: Human-guided Robot Object Search
- Committee: Stefanie Tellex (advisor), Ellie Pavlick, Srinath Sridhar
- Master of Science in Computer Science (May 2020)

2014–2018 **Vassar College**, Bachelor of Arts in Computer Science & Minor in Economics.
- 3.9/4.0 GPA. Received General Honors and Honors in Computer Science

Academic Positions

2024–Present **Assistant Professor of Computer Science**, Haverford College.
- Faculty Pedagogy Seminar & Students as Learners and Teachers program (Fall 2024), *TLI*

2018–2024 **Graduate Research Assistant**, *Humans to Robots Laboratory*, Brown University.

Fall 2023 **Lead Lecturer**, *Artificial Intelligence (course size: 115)*, Brown University.

Spring 2021 **Graduate Teaching Assistant**, *Topics in Collaborative Robotics (17)*, Brown University.

2016–2018 **Research Assistant**, Vassar College.

Fall 2017 **Head Teaching Assistant**, *Advanced Topics: Intelligent Robotics (5)*, Vassar College.

Summer 2016 & 2017 **Fellow**, *Undergraduate Research Summer Institute*, Vassar College.

Spring 2016 **Teaching Assistant**, *Foundations of Computer Science (40)*, Vassar College.

Publications & Preprints

*Equal contribution, ˆUndergraduate student

Journal Papers

AURO 2022 **Affordance-based robot object retrieval**
Thao Nguyen, Nakul Gopalan, Roma Patel, Matt Corsaro, Ellie Pavlick, Stefanie Tellex

AURO 2022 **Hierarchical planning with state abstractions for temporal task specifications**
Yoonseon Oh, Roma Patel, Thao Nguyen, Baichuan Huang, Matthew Bergˆ, Ellie Pavlick, Stefanie Tellex

Conference Papers

CogSci 2024 **Find it like a dog: Using Gesture to Improve Object Search**
Acceptance Rate: 19%
Madeline H. Pelgrim, Ivy Xiao He, Kyle Leeˆ, Falak Pabariˆ, Stefanie Tellex, Thao Nguyen, Daphna Buchsbaum

IROS 2023 **Language-Conditioned Observation Models for Visual Object Search**
AR: 43%
Thao Nguyen, Vladislav Hrosinkovˆ, Eric Rosen, Stefanie Tellex

RSS 2020 **Robot Object Retrieval with Contextual Natural Language Queries**
AR: 32%
Thao Nguyen, Nakul Gopalan, Roma Patel, Matt Corsaro, Ellie Pavlick, Stefanie Tellex

IROS 2019 **Grounding Language Attributes to Objects using Bayesian Eigenobjects**
AR: 43% Vanya Cohen*, Benjamin Burchfiel*, Thao Nguyen*, Nakul Gopalan, Stefanie Tellex, George Konidaris

RSS 2019 **Planning with State Abstractions for Non-Markovian Task Specifications**
AR: 31% Yoonseon Oh, Roma Patel, Thao Nguyen, Baichuan Huang, Ellie Pavlick, Stefanie Tellex

Workshop Papers

CoRL 2026 (under review) **Structure over Latent Maps: A Neuro-Symbolic World Model for Interactive Object Navigation**
Owen Bluman^, Maxfield Ma^, Thao Nguyen

IROS 2026 **Meanings and Measurements: Multi-Agent Probabilistic Grounding for Vision-Language Navigation**
Lakshya Jain, Swagat Padhan, Bhavya Minesh Shah, Omkar Patil, Thao Nguyen, Nakul Gopalan

RSS 2022 **Towards Language-Conditioned Observation Models for Visual Object Search**
Thao Nguyen, Vladislav Hrosinkov^, Eric Rosen, Stefanie Tellex

In Preparation

2026 **Improving VLMs for Object Identification Under Partial Occlusion with Synthetically Occluded Images**
Ryan Sam Varghese, Weiwei Gu, Thao Nguyen, Ifrah Idrees, Nakul Gopalan

HDSR 2026 **Generative AI hasn't changed learning: It's still hard to become an expert**
Ming-Wen An, Benjamin S. Baumer, Logan Dancey, Richard De Veaux, Daryl DeFord, Erika Franklin Fowler, Stephen Freund, Sorelle Friedler, Maryam Gooyabadi, Douglas Hall, Brianna Heggeseth, Nicholas J. Horton, Emmanuel Kaparakis, Antonio Laverghetta Jr., Victoria Manfredi, Valerie Nazzaro, Thao Nguyen, Cassandra Pattanayak, Anna Plantinga, Alex Reinhart, Lynne Steuerle Schofield, Marc Schulz, Sarah Supp, Anjali Thapar, Elizabeth Upton, Zhe Wang

Teaching & Mentoring

Teaching

CMSC369 **Introduction to Robotics**, Spring 2026 *(course size: 19)*.
- New course added to the Haverford curriculum

CMSC260 **Foundations of Data Science**, Fall 2024 *(24)*, Fall 2025 *(36)*, Fall 2026 *(17)*.

CMSC105 **Introduction to Computer Science**, Spring 2025 *(15)*.

Mentoring

AY 26-27 **Senior Thesis Mentees**.
- Owen Bluman '27
- Nicole Huang '27
- Maxfield Ma '27
- Ken Shen '27
- August Taylor '27
- Mateo Taylor '27

2026 **Summer Student Mentees**.
- Owen Bluman '27
- Maxfield Ma '27
- Ryaan Sharif '28

AY 25-26 **Senior Thesis Mentees**.

- Conrad Coyanda-Parkzes '26, *Use Cases of LLMs in Robot-Human Interaction*
- Meghan Galban '26, *Eliciting User Preference Through Dialog and Query Generation With Robots For Object Search*
- Aiden Kim '26, *Visual Concept Learning with Robots*
- Rami Nasr '26, *Clarifying Questions in Robotics: Methods for Object Disambiguation*
- Jaxon Remby '26, *Enhancing LLM-Based Robotic Search through Semantic Room Annotation*

2025 **Summer Student Mentees**.

- Owen Bluman '27
- Zenas Boamah '27
- Matt Feinstein '27
- Lei Lei '26

AY 24-25 **Senior Thesis Mentees**.

- Jennifer Contreras-Ortiz '25, *Exploring Hallucination Mitigation Techniques Within a Question-Answering Framework*
- Edgar Leon '25, *Reducing Human Oversight in Robotic Learning: Predicate-Based Approaches with Generative Language Models*
- Jonathan Moon '25, *Recounting Robots: Leveraging Multimodal Data for Improved Transparency and Task Performance*
- Daniel Rosman '25, *Success Detection in Robotics: Action Success vs Failure*
- Evan Wolpert '25, *Uncertainty Quantification and Communication in Large Language Models*

Service

Professional Service

HRI 2026 & 2027 **Conference Reviewer**.

THRI 2024 & 2026 **Journal Reviewer**.

IROS 2026 **Organizer**, Workshop on Bridging the Gap between Neural and Symbolic World Models for Robot Planning, Reasoning, and Action.

RSS 2024 **Conference Reviewer**.

ICRA 2021–2024 **Conference Reviewer**.

IROS 2023 **Conference Reviewer**.

AAAI-22 **Graduate Student Fellow & Panel Speaker**, Undergraduate Consortium.

EMNLP 2021 & 2022 **Program Committee**, Novel Ideas in Learning-to-Learn through Interaction Workshop.

AAAI-21 **Invited Speaker**, Graduate Student Panel, Undergraduate Consortium.

Fall 2020 **Mentor**, Brown Computer Science PhD Application Support for Underrepresented Students.

Haverford College

Fall 2026 – Present **Department Member Representative**, Computer Science Tenure-track Faculty Search Committee.

Fall 2024 – Present **Faculty-Student Liaison & Event Planning Committee Member**, Computer Science Department.

Spring 2026 **Department Representative**, Computer Science Visiting Faculty Search Committee.

AY 25-26 **Faculty Representative**, Advisory Committee, Haverford Innovations & Entrepreneurship Program.

Invited Talks

Jan 2024 **Human-guided Robot Object Search**, Haverford College.

Conference Presentations

Oct 2023 **Language-Conditioned Observation Models for Visual Object Search**, IEEE/RSJ International Conference on Intelligent Robots and Systems, Detroit, MI, USA.

July 2020 **Robot Object Retrieval with Contextual Natural Language Queries**, Robotics: Science and Systems, held virtually.

Guest Lectures

Nov 2022 **Robotics**, *Artificial Intelligence*, Brown University.

Nov 2022 **Natural Language Processing**, *Artificial Intelligence*, Brown University.

Awards

2025 **Haverford Teaching with Technology Award**, $4800.

2021 **Scholar**, CRA-WP Grad Cohort for Women.

2016 **Scholar**, Grace Hopper Celebration of Women in Computing.